

FREE EBOOK

PHASE 1 QUANT & REASONING FORMULAE E-BOOK



FOR RBI GRADE B EXAM

Phase 1 Quant & Reasoning Formulae for RBI Grade B

The RBI Grade B Phase 1 exam is designed to evaluate a candidate's speed, accuracy, and analytical thinking through the Quantitative Aptitude and Reasoning sections. These sections demand a strong grasp of core concepts and the ability to apply formulas quickly under time pressure. A solid command over essential formulas not only boosts calculation speed but also reduces errors, giving you a competitive edge. This eBook compiles all the important formulas and concepts from both Quant and Reasoning topics in a chapter-wise format, making it perfect for quick revision and last-minute prep.

Quant Formulas

This table summarizes key formulas used in quantitative aptitude topics like percentages, profit & loss, simple interest, time & work, geometry, algebra, and more. It serves as a quick reference guide for students preparing for competitive exams or brushing up on math fundamentals.

Topic	Formula	Description
Percentage	$\% \text{ Increase} = \frac{(\text{New} - \text{Old})}{\text{Old}} \times 100$	To calculate percentage increase
	$\% \text{ Decrease} = \frac{(\text{Old} - \text{New})}{\text{Old}} \times 100$	To calculate percentage decrease
	$\text{Population after } n \text{ years} = P(1 + R/100)^n$	For population growth
	$\text{Depreciation} = P(1 - R/100)^n$	Value decreases over time
Profit & Loss	$\text{SP} = \frac{(100 + \text{Profit}\%)}{100} \times \text{CP}$	Find selling price when profit % is given
	$\text{SP} = \frac{(100 - \text{Loss}\%)}{100} \times \text{CP}$	Find selling price when loss % is given
	$\text{CP} = \frac{100}{(100 + \text{Profit}\%)} \times \text{SP}$	Find cost price from selling price and profit %
Simple Interest	$\text{SI} = \frac{\text{PTR}}{100}$	P = Principal, T = Time, R = Rate
	$\text{Total Amount} = \text{SI} + \text{P}$	Final amount received
Compound Interest	$\text{CI (Annual)} = P[(1 + R/100)^n - 1]$	CI over n years annually compounded
	$\text{CI (Half-Yearly)} = P[(1 + R/200)^{(2n)} - 1]$	Compounded semi-annually

	$CI \text{ (Quarterly)} = P[(1 + R/400)^{(4n)} - 1]$	Compounded quarterly
Time & Distance	Relative Speed (Same Direction) = (faster - slower)	For two moving objects in same direction
	Relative Speed (Opposite Direction) = (faster + slower)	For objects in opposite direction
Trains	Time = Length / Speed	Time taken to cross object
	Time to cross platform = (Train length + Platform length)/Speed	Applies to stationary platforms
	Time to cross man = Train length / Speed	Man assumed stationary
Boats & Streams	Time = Distance / Speed	Time formula still applies
	Total time = Distance/Downstream + Distance/Upstream	If round trip is involved
Time & Work	Work = Time × Efficiency	Work is product of time and efficiency
	Efficiency = 1 / Time	Inverse of time
	If A is x times as good as B, Ratio of work = x:1	Useful in comparative efficiency
Pipes & Cisterns	Net Work = (Inlet rate - Outlet rate)	For combined pipe flows
	Time to empty when both run = 1 / (outlet rate - inlet rate)	When outlet is faster
Averages	Average = (Sum of n items)/n	Basic average
	New Average = (Old Sum ± Change) / n	When a value is added or removed
Ratio & Proportion	Inverse Ratio = $1/a : 1/b = b : a$	Inverse proportion
	Mean Proportion = $\sqrt{a \times b}$	Middle term between two values
	Third Proportion = b^2 / a	If $a:b = b:c$, c is third proportion
Mixtures	Alligation Rule: Quantity Ratio = (CP1 - Mean):(Mean - CP2)	CP = Cost Price
	Final Concentration = $(xA + yB) / (x + y)$	Mixing two solutions

Mensuration (2D)	Perimeter of Rectangle = $2(l + b)$	Boundary length
	Perimeter of Square = $4a$	Side $\times 4$
	Circumference of Circle = $2\pi r$	Circular boundary
	Area of Rhombus = $(\frac{1}{2}) \times d_1 \times d_2$	d_1, d_2 = diagonals
Mensuration (3D)	TSA (Cube) = $6a^2$	Total Surface Area
	TSA (Cuboid) = $2(lb + bh + hl)$	Total surface of cuboid
	CSA (Cylinder) = $2\pi rh$	Curved Surface Area
	TSA (Cylinder) = $2\pi r(h + r)$	Includes top & bottom
	Volume of Cone = $(\frac{1}{3})\pi r^2 h$	Volume of cone
	Volume of Sphere = $(\frac{4}{3})\pi r^3$	Volume of sphere
	Surface Area of Sphere = $4\pi r^2$	Surface of sphere
Algebra	$(a + b)^2 = a^2 + 2ab + b^2$	Expansion formula
	$(a - b)^2 = a^2 - 2ab + b^2$	Expansion formula
	$a^2 - b^2 = (a + b)(a - b)$	Identity formula
	$(x + a)(x + b) = x^2 + (a + b)x + ab$	Binomial product
Number Series	n th term of AP = $a + (n - 1)d$	Arithmetic Progression
	Sum of n terms of AP = $\frac{n}{2} \times [2a + (n - 1)d]$	Sum of AP terms
	n th term of GP = $a \times r^{(n - 1)}$	Geometric Progression
	Sum of n terms of GP = $a[(r^n - 1)/(r - 1)]$	$r \neq 1$
Permutation & Comb.	$nP_n = n!$	All items arranged
	$nCr = n! / (r!(n - r)!)$	r items selected from n
Probability	$P(E) = \text{Favourable Outcomes} / \text{Total Outcomes}$	Fundamental formula
	$P(A \cap B) = P(A) \times P(B)$	For independent events
	$P(\text{not } A) = 1 - P(A)$	Complement rule

Reasoning Formulas

This table summarizes important concepts and shortcut techniques used in reasoning topics such as puzzles, seating arrangements, syllogisms, inequalities, coding-decoding, blood relations, and more. It serves as a quick reference guide for aspirants preparing for competitive exams like RBI Grade B, helping them revise logic-based patterns, decision-making approaches, and time-saving strategies efficiently.

Section	Chapter	Topic	Key Formulae/Concepts
Quantitative Aptitude	1	Number System	Divisibility, LCM-HCF, Euler's & Fermat's Theorem, Factorials
	2	Simplification & Approximation	Surds, Indices, BODMAS, Common Approximations
	3	Percentage, Profit & Loss	% Change, Successive Change, MP-SP-CP Relationships
	4	SI & CI	$SI = \frac{PRT}{100}$, $CI = P(1 + \frac{R}{100})^T$, Effective Rate
	5	Ratio & Proportion, Partnership	Ratio Basics, Profit Sharing, Time-Joined Partnership
	6	Time, Speed & Distance	$S = \frac{D}{T}$, Relative Speed, Boats & Streams
	7	Time & Work	Work = Rate $\times$ Time, Efficiency Ratios, Pipes & Cisterns
	8	Algebra	Algebraic Identities, Quadratic Formula, Inequalities
	9	Mensuration	2D & 3D Figures Area/Volume, Frustum, Hemisphere
	10	Data Interpretation	% Change, Average, Approximation, DI Tricks
Reasoning Ability	11	Seating & Puzzle	Linear, Circular, Floor & Box-based, Tabular Layout
	12	Syllogism	A-E-I-O Rules, Venn Diagram, Reverse Syllogism
	13	Inequality	Direct, Coded, Transitive, Symbol Reversal
	14	Coding-Decoding	Letter Shifting, Number Coding, Fictitious Language
	15	Blood Relations	Tree Mapping, Puzzle-based, Coded Relations
	16	Direction Sense	Turns, Distance = $\sqrt{x^2 + y^2}$, Map Drawing
	17	Input-Output	Pattern Recognition, Word-Number Logic
	18	Logical Reasoning	Assumptions, Conclusions, Cause-Effect



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